INCH-POUND

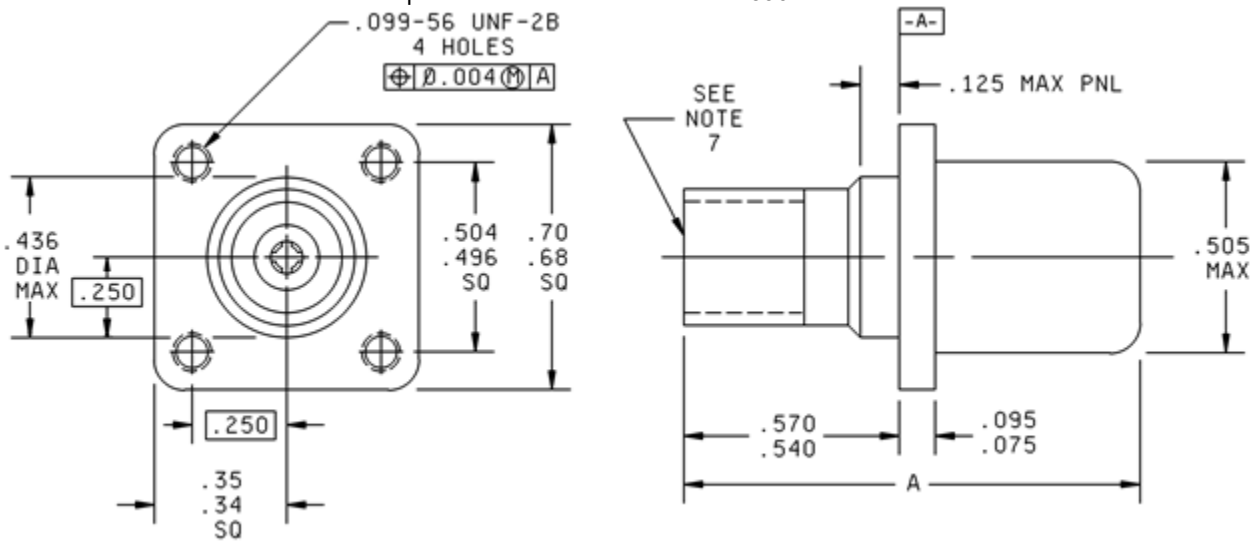
MIL-PRF-39012/29H
w/AMENDMENT 4
27 February 2020
SUPERSEDING
MIL-PRF-39012/29H
w/AMENDMENT 3
25 January 2018

PERFORMANCE SPECIFICATION SHEET

# CONNECTORS, RECEPTACLE, ELECTRICAL, COAXIAL, RADIO FREQUENCY, (SERIES TNC, (CABLED), SOCKET CONTACT, FLANGE MOUNTED, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39012.



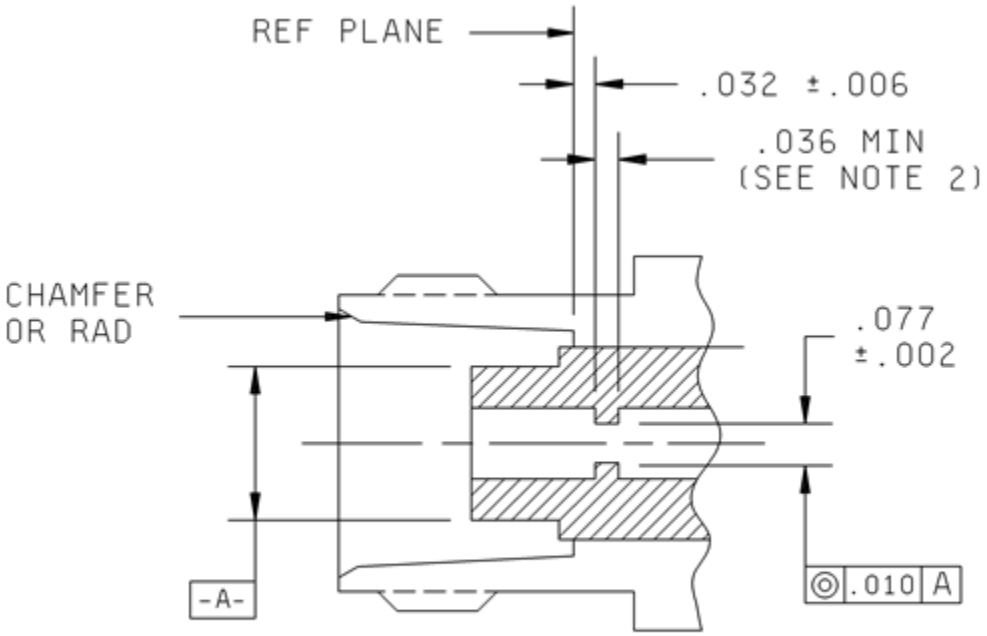

| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .004 | 0.10 | .34 | 8.64 | .540 | 13.72 |
| .075 | 1.90 | .35 | 8.89 | .570 | 14.48 |
| .095 | 2.41 | .436 | 11.07 | .68 | 17.27 |
| .099 | 2.51 | .496 | 12.60 | .70 | 17.78 |
| .125 | 3.18 | .504 | 12.80 | | |
| .250 | 6.35 | .505 | 12.83 | | |

NOTES:
1. Dimensions are in inches. Metric equivalents are given for information only.
2. For dimension A, see tables I and III.
3. Wrench flats (if required for assembly) are to accommodate standard wrench opening in accordance with FED-STD-H28.
4. All undimensioned pictorial representations are for reference purposes only.
5. Receptacle not recommended for use on panels with greater than .078 (1.98 mm) maximum thickness.
6. Series TNC, socket contact interface in accordance with MIL-STD-348.
7. Dimension A defines the maximum length of the connector when assembled to the appropriate cable.

FIGURE 1. General configuration.



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CATEGORY D

| Inches | mm |
|---|---|
| .002 | 0.05 |
| .006 | 0.15 |
| .010 | 0.25 |
| .032 | 0.81 |
| .036 | 0.91 |
| .077 | 1.96 |

NOTES:
1. Dimensions are in inches. Metric equivalents are given for information only.
2. Chamfer is optional.
3. All undimensioned pictorial configurations are for reference purposes only.
4. Interface in accordance with MIL-STD-348 series TNC, socket contact.

FIGURE 2. Category D captivation detail.



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TABLE I. Dash numbers, cross-reference, and dimensions.

| Part or Identifying Number (PIN) 1/ M39012/29- | Applicable cable 2/ M17/ | Dimensions | Inch (millimeters) maximum # |
|---|---|---|---|
| Category A – Field serviceable (no special tools required) 3/ | | | |
| X101 | Cable group VI 6/<br>60-RG142 4/<br>128-RG400 5/ | A | 1.250 (31.75) |
| X102 | Cable group VII 6/<br>110-RG302 4/ 5/ | | |
| X103 | Cable group IV 6/<br>54-RG122 5/ | | |
| X018 | Cable group II 6/<br>113-RG316 4/ 5/ | | |
| X225 | Cable group X 6/<br>127-RG393 4/ 5/ | | |
| 0030 | 220-00001 | | |
| 0130 | 220-00002 13/ | | |
| 0031 | 221-00001 | | |
| 0131 | 221-00002 13/ | | |
| 0032 | 222-00001 | | |
| 0132 | 222-00002 13/ | | |
| 0033 | 223-00001 | | |
| 0133 | 223-00002 13/ | | |
| 0034 | 224-00001 | | |
| 0134 | 224-00002 13/ | | |
| 0035 | 225-00001 | | |
| 0135 | 225-00002 13/ | | |

See notes at end of table.



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TABLE I. Dash numbers, cross-reference, and dimensions – Continued.

| PIN 1/<br>M39012/29- | Applicable cable 2/<br>M17/ | Dimensions | Inches (millimeters)<br>maximum # |
|---|---|---|---|
| Category C – field replaceable (SAE-AS22520/5-01 crimp tool)<br>See note next to applicable cable for crimp die 3/ 7/ | | | |
| X010 | Cable group VIA 6/ 8/<br>111-RG303 4/ 5/ | A | 1.500 (38.10) |
| X011 | Cable group VIB 6/ 8/<br>60-RG142 4/<br>128-RG400 5/ | | |
| X012 | Cable group VIIA 6/ 9/<br>110-RG302 4/ 5/ | | |
| X013 | Cable group IV 6/ 10/<br>54-RG122 5/ | | |
| X014 | Cable group VIIB 6/ 9/<br>90-RG71 5/ | | |
| X022 | Cable group IIA 6/ 11/<br>113-RG316 4/ 5/ | | |
| X023 | Cable group V 6/ 10/<br>95-RG180 5/ | | |
| Category D – Field replaceable – Defined piece part 3/ 7/ 12/ | | | |
| X501 | Cable group IV 6/<br>54-RG122 5/ | A | 1.468 (37.31) |

See notes at end of table.



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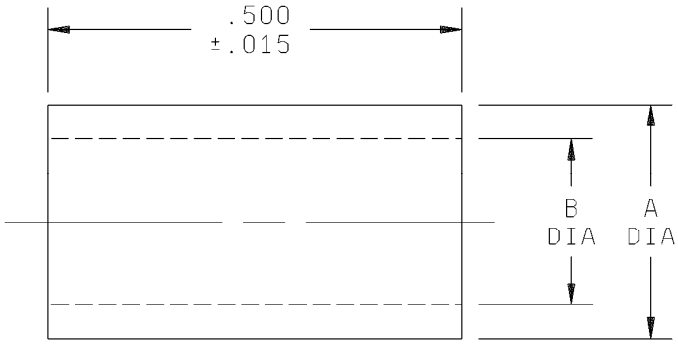
TABLE I. Dash numbers, cross-reference, and dimensions – Continued.

| PIN 1/ M39012/29- | Applicable cable 2/ M17/ | Dimensions | Inches (millimeters) maximum # |
|---|---|---|---|
| Category D – Field replaceable – Defined piece part 3/ 7/ 12/ | | | |
| X502 | Cable group V 6/<br><br>95-RG180 4/ 5/ | A | 1.234 (31.35) |
| X503 | Cable group VIB 6/<br><br>60-RG142 4/<br>128-00001 5/ | | |
| X504 | Cable group VIA 6/<br><br>111-RG303 4/ 5/ | | |

1/ For cross-reference of PIN to superseded PIN or designation, see table IV.
2/ The latest version of each cable shall be applicable.
3/ These connectors have captivated center contacts.
4/ Cable to be used for the +200°C temperature cycling tests. This cable can be used for tests with the approval of the Qualifying Activity.
5/ Cable to be used when performing test requiring cable except as in 5/ and 7/.
6/ Cables that are not 50-ohm cables in this cable group, when attached to the specified connectors, VSWR, RF leakage and insertion loss are not applicable.
7/ These connectors are assembled using the applicable crimp tool, to the specified cables stripped as shown on figure 4.
8/ M22520/5-19 closure B or M22520/5-05 closure A. M22520/5-11 closure A. M22520/5-57 closure A.
9/ M22520/5-19 closure A or M22520/5-07 closure A. M22520/5-13 closure A. M22520/5-59 closure A.
10/ M22520/5-41 closure B or M22520/5-05 closure B. M22520/5-09 closure A.
11/ M22520/5-35 closure B or M22520/5-03 closure A.
12/ Complete connector assembly shall consist of a body, center contact, ferrule and assembly instructions.
13/ Armored cable.
# Dimensions are in inches. Metric equivalents are given for information only.
X Denotes connector body plating material option. The only plating options allowable are Silver or Nickel over brass in accordance with MIL-PRF-39012. Only connectors of the same materials shall be mated to avoid dissimilar metal problems. **CAUTION: A NICKEL PLATED BODY IS NOT FOR USE IN APPLICATIONS WHERE PASSIVE INTERMODULATION GENERATION (PIM) MAY BE A CONCERN.** Silver is the preferred plating option.



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CRIMP FERRULE

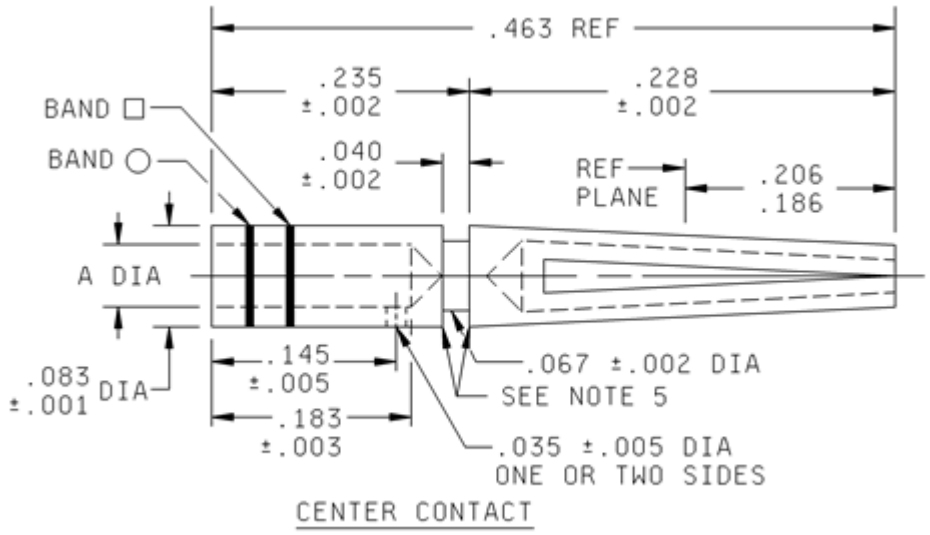
| Dash number | Ferrule number 1/ | A ± 0.003 | B ± 0.003 | Basic crimp tool 2/ | Crimp die or positioner M22520/5 |
|---|---|---|---|---|---|
| X501<br>X502 | 29-50 | 0.212 | 0.175 | M22520/5-01 | 05, 41 Closure B or 9 Closure A |
| X503 | 29-51 | 0.250 | 0.220 | | 05, 11, 57 Closure A or 19 closure B |
| X504 | 29-52 | 0.245 | 0.206 | | |

1/ Contact numbers and ferrule numbers are for identification only.
2/ Class 2 tool may be used by OEM (see SAE-AS22520).

FIGURE 3. Contact and ferrule dimensions for category D only.



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CENTER CONTACT

| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .001 | 0.03 | .033 | 0.84 | .145 | 3.68 |
| ,002 | 0.05 | .035 | 0.89 | .175 | 4.45 |
| .003 | 0.08 | .040 | 1.02 | .183 | 4.65 |
| .005 | 0.13 | .043 | 1.09 | .206 | 5.23 |
| .015 | 0.38 | .067 | 1.70 | .228 | 5.79 |
| .017 | 0.43 | .083 | 2.11 | .235 | 5.97 |
| | | | | .463 | 11.76 |

| Dash No. | Contact No. 1/ | A +0.001 -0.002 | Basic crimp tool 2/ | Crimp die or positioner | Crimp tool setting | Crimp Tensile min. (N) | Color band □ | Color band ○ |
|---|---|---|---|---|---|---|---|---|
| X501 | 29-12 | 0.033 | M22520/1-01 | M22520/1-12 | 0.033 | 10 lbs. (44.48) | Orn | Green |
| X502 | 29-11 | 0.017 | | | 0.017 | 6 lbs. (26.69) | Blue | |
| X503 X504 | 29-10 | 0.043 | M22520/1-01 | M22520/1-12 | 0.043 | 20 lbs. (88.96) | Red | |

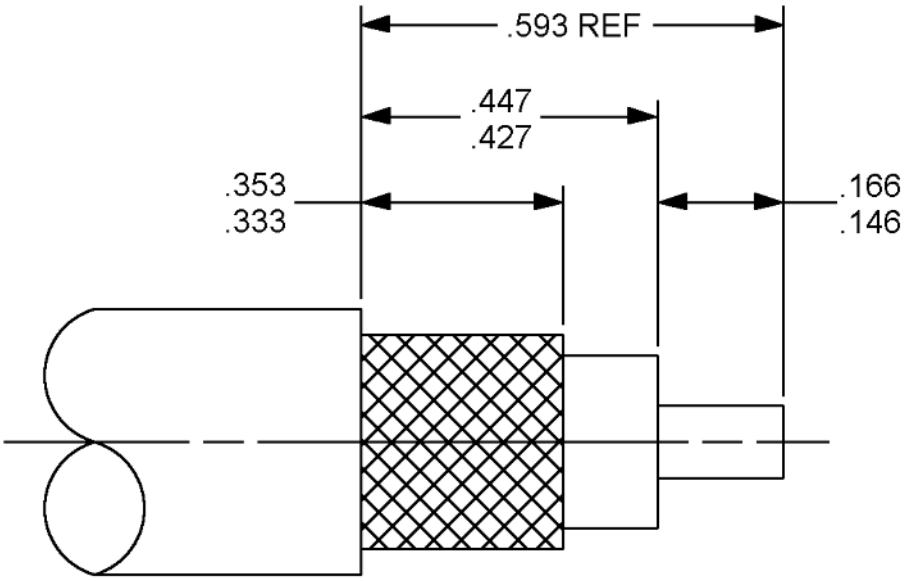
1/ Contact numbers and ferrule numbers are for identification only.
2/ Class 2 tool may be used by OEM (see SAE-AS22520).

NOTES:
1. Dimensions are in inches. Metric equivalents are given for information only.
2. Contact material shall be copper beryllium. Connectors supplied with phosphor bronze contacts are no longer acceptable for Government use.
3. Crimp tensile test shall be in accordance with SAE-AS39029.
4. Copyright notice: All information disclosed in these specification sheets which is or may be copyright is reproduced herein with the express permission of the copyright owner.
5. .003 inch maximum break.
6. Color bands shall be positioned so that no coloring material enters the inspection hole.

FIGURE 3. Contact and ferrule dimensions for category D only – Continued.



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| Inches | mm |
|---|---|
| .146 | 3.71 |
| .166 | 4.22 |
| .333 | 8.46 |
| .353 | 8.97 |
| .427 | 10.85 |
| .447 | 11.35 |
| .593 | 15.06 |

NOTES:
1. Dimensions are in inches. Metric equivalents are given for Information only.

FIGURE 4. Cable stripping dimensions for field replaceable connectors.



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ENGINEERING DATA

Nominal impedance: 50 ohms.

Frequency range: 0 to 11 GHz.

Voltage rating:

500 volts rms, maximum working voltage at sea level.

125 volts rms, maximum at 70,000 feet (4.437 kPa).

Temperature rating: - 65°C to + 165°C.

REQUIREMENTS

Designs and configurations: See figures 1, 2, 3, and 4.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque – 2 inch-pounds (.22 Nm), maximum.

Coupling proof torque: Not applicable.

Inspection conditions:

Coupling torque: 4 – 6 inch-pounds ((0.51 Nm to 0.67 Nm).

Mating characteristics:

In accordance with MIL-STD-348 and figure 2 for dimensions.

Center contact (socket):

Oversize test pin: .057 inch (1.45 mm) diameter minimum (nonclosed entry contacts only).

Insertion depth: .125 inch (3.17 mm) minimum.

Number of insertions: 1.

Insertion force test: Steel test pin diameter .054 inch (1.37 mm) minimum.

Test pin finish: 16 microinches (4.437 kPa).

Insertion force: 2 pounds (8.90 N), maximum.



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Withdrawal force test: Steel test pin diameter .052 inch (1.32 mm), maximum.

Withdrawal force: 2 ounces (.56 N), minimum.

Test pin finish: 16 microinches (0.406 µm).

Hermetic seal: Not applicable.

Leakage: Not applicable.

Air pressure: 30 Lbf/in$^2$.

Duration: 30 seconds, minimum.

Insulation resistance: MIL-STD-202-302, test condition B. 5,000 megohms minimum.

Center contact retention: 6 pounds (26.69 N) minimum axial force. Applicable to captivated-center-contact connectors only.

Corrosion (salt spray): MIL-STD-202-101, test condition B.

Voltage standing wave ratio (VSWR): From .5 to 11 GHz, or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower; 1.30 maximum.

Swept frequency VSWR test setup:

Item 6 - VSWR shall be less than 1.02 + .003 F (F in GHz).

Item 16- VSWR shall be less than 1.02 + .003 F (F in GHz).

Second step of VSWR checkout procedure - VSWR shall be less than 1.06 + .007 F (F in GHz).

Group B inspection - VSWR shall be less than 1.05 + .005 F (F in GHz).

Qualification and group C inspection - VSWR shall not exceed 1.10.

Connector durability: 500 cycles minimum at 12 cycles/minute maximum. The connector shall meet the mating characteristics and force to engage end disengage requirements.

Contact resistance: In milliohms maximum:

| | Initial | After environment |
|---|---|---|
| Center contact | 1.5 | 2.0 |
| Outer contact (silver) | .2 | Not applicable |
| Outer contact (nickel) | .4 | Not applicable |
| Braid to body | .1 | Not applicable |

Dielectric withstanding voltage: MIL-STD-202-301, 1,500 volts rms minimum at sea level.

Vibration, high frequency: MIL-STD-202-204, test condition B.

Shock: MIL-STD-202-213, test condition I.



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Thermal shock: MIL-STD-202-107, test condition B, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200° cables (see tables I and III).

Moisture resistance: MIL-STD-202-106. No measurements at high humidity. Insulation resistance shall be at least 200 megaohms within 5 minutes after removal from humidity.

Corona Level:

Voltage - 375 volts rms, minimum.

Altitude - 70,000 feet (4.437 kPa).

RF high potential withstanding voltage:

Voltage and frequency: 1,000 volts rms at a frequency from 5 to 7.5 MHz.

Leakage current: Not applicable.

Cable retention force:

Noncrimp assemblies: 40 pounds (177.92 N), minimum.

Crimp assemblies:

10 pounds (44.48 N), minimum for cables .089- .189 inch (2.26 mm – 4.80 mm) OD.

20 pounds ((88.96 N), minimum for cables .190-.229 inch (4.83 mm – 5.82 mm) OD.

30 pounds (133.45 N), minimum for cables .230-.249 inch (5.84 mm – 6.32 mm) OD.

40 pounds (177.92 N), minimum for cables .250 inch (6.35 mm) OD. and larger.

Coupling mechanism retention force: Not applicable.

RF Leakage: -60 dB minimum, tested at a frequency between 2 and 3 GHz.

Insertion loss: .18 dB maximum tested at 9 GHz.

$.06\sqrt{F\ (GHz)}$ dB max tested at 3 GHz to 6 GHz.

PIN: M39012/29 - (dash number from table I or “B” number from table III).

Group qualification: See table II.



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TABLE II. Group qualification. 1/

| Group | Submission and qualification of 2/ any of the following connectors | Qualifies the following connectors |
|---|---|---|
| I | M39012/<br><br>27-X101<br>27-X103<br>27-X018<br>28-X101<br>28-X103<br>28-X018<br>29-X101<br>29-X103<br>29-X018 | M39012/<br><br>27-X101<br>27-X002<br>27-X103<br>27-X117<br>27-X018<br>28-X101<br>28-X102<br>28-X103<br>28-X117<br>28-X018<br>29-X101<br>29-X102<br>29-X103<br>29-X117<br>29-X018 |
| II | 27-X002<br>28-X102<br>29-X102 | 27-X002<br>28-X102<br>29-X102 |
| III | 27B0005<br>27B0006<br>27B0008<br>27B0015<br>27B0016<br>27B0020<br>28B0005<br>28B0006<br>28B0008<br>28B0015<br>28B0016<br>28B0020<br>29B0005<br>29B0006<br>29B0008<br>29B0015<br>29B0016<br>29B0020 | 27B0005<br>27B0006<br>27B0007<br>27B0008<br>27B0009<br>27B0015<br>27B0016<br>27B0019<br>27B0020<br>28B0005<br>28B0006<br>28B0007<br>28B0008<br>28B0009<br>28B0015<br>28B0016<br>28B0019<br>28B0020 |

See notes at end of table.



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.TABLE II. Group qualification – Continued. 1/

| Group | Submission and qualification of 2/ any of the following connectors | Qualifies the following connectors |
|---|---|---|
| III | | 29B0005<br>29B0006<br>29B0007<br>29B0008<br>29B0009<br>29B0015<br>29B0016<br>29B0019<br>29B0020 |
| IV | 27B0007<br>27B0009<br>27B0019<br>28B0007<br>28B0009<br>28B0019<br>29B0007<br>29B0009<br>29B0019 | 27B0007<br>27B0009<br>27B0019<br>28B0007<br>28B0009<br>28B0019<br>29B0007<br>29B0009<br>29B0019 |
| V | 27-X010<br>27-X011<br>27-X013<br>27-X022<br>28-X010<br>28-X011<br>28-X013<br>28-X022<br>29-X010<br>29-X011<br>29-X013<br>29-X022 | 27-X010<br>27-X011<br>27-X012<br>27-X013<br>27-X014<br>27-X022<br>28-X010<br>28-X011<br>28-X012<br>28-X013<br>28-X014<br>28-X022<br>29-X010<br>29-X011<br>29-X012<br>29-X013<br>29-X014<br>29-X022 |

See notes at end of table.



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TABLE II. Group qualification – Continued. 1/

| Group | Submission and qualification of any of the following connectors 2/ | Qualifies the following connectors |
|---|---|---|
| VI | 27-X012<br>27-X014<br>27-X021<br>28-X012<br>28-X014<br>29-X012<br>29-X014 | 27-X012<br>27-X014<br>28-X012<br>28-X014<br>29-X012<br>29-X014 |
| VII | 29-X501<br>29-X503<br>29-X504 | 29-X501<br>29-X502<br>29-X503<br>29-X504 |
| VIII | 29-X502 | 29-X502 |

1/ If a connector manufacturer produces a connector which meets all the requirements for two or more connector PINs (within the same series), the manufacturer may receive qualification approval for two or more connector PINs qualifying the one connecter. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate PIN. For group qualification, the connectors must be of similar design and be of the same materials and plating.

2/ For qualification retention, where more than one part is listed in a group in this column, data may be supplied on any of those parts in order to retain qualification for those parts in the corresponding right hand column. The part does not necessarily have to be the part initially qualified.



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TABLE III. Category B – non-field replaceable (special tools may be required). 1/

Not for Air Force, Army, or Navy use. For OEM use only.

| PIN 2/ 3/ M39012/29B | Applicable Cable M17/ 4/ | Dimensions | Inches (millimeters) maximum # |
|---|---|---|---|
| 0005 | 28-RG058 5/ | A | 1.500 (38.10) |
| 0006 | 060-RG142 5/ 7/<br>128-RG400 | | |
| 0007 | 29-RG59 5/ 6/<br>030-RG062<br>97-RG210 | | |
| 0008 | 54-RG122 5/ | | |
| 0009 | 90-RG71 5/ 6/ | | |
| 0015 | 111-RG303 5/ 7/ | | |
| 0016 | 084-RG223 5/ | | |
| 0019 | 110-RG302 5/ 6/ 7/ | | |
| 0020 | 113-RG316 5/ 7/<br>119-RG174 | | |

1/ For maintenance replacements for category B, see table V.
2/ For cross-reference of PIN to superseded PIN or type designation, see table IV.
3/ Inactive for new design.
4/ The latest version of each cable shall, be applicable.
5/ Cable to be used when performing tests requiring cable except as in 6/ and 7/.
6/ These are not 50 ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.
7/ Cable to be used for the +200°C temperature cycling tests. This cable can be used for tests with the approval of the Qualifying Activity.
# Dimensions are in inches. Metric equivalents are given for information only.



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TABLE IV. Supersession data. 1/ 2/

| Preferred PIN M39012/29 | Superseded PINs or superseded type designation |
|---|---|
| -0012 | M39012/29-0021 |
| -0101 | M39012/29-0001 - M39012/29-0004, M39012/29-0104 |
| -0102 | M39012/29-0002 - M39012/29-0017, M39012/29-0117 |
| -0103 | M39012/29-0003 |
| B0005 | M39012/29-0005 |
| B0006 | M39012/29-0006 |
| B0007 | M39012/29-0007 |
| B0008 | M39012/29-0008 |
| B0009 | M39012/29-0009 |
| B0015 | M39012/29-0015 |
| B0016 | M39012/29-0016 |
| B0019 | M39012/29-0019 |
| B0020 | M39012/29-0020 |

1/ The superseded PIN or the type designation is for reference only. Where a superseded PIN or type designation is not given, none was assigned or will be assigned. The PIN M39012/29-XXXX shall be used in all cases for marking and identifying the connector.

2/ The basic type designation includes all letter versions of the specified number, e.g. UG-18/U includes UG-18 A/U, UG-18B/U, etc.

TABLE V. Maintenance replacements for category B.

| Category B number Inactive for new design 1/ | Category C dash number | Category A dash number | Category D dash number |
|---|---|---|---|
| B0005 | 0010 | 0101 | 0504 |
| B0006 | 0011 | 0101 | 0503 |
| B0007 | 0012 | 0102 | --- |
| B0008 | 0013 | 0103 | 0501 |
| B0009 | 0014 | 0102 | --- |
| B0015 | 0010 | 0101 | --- |
| B0016 | 0011 | 0101 | --- |
| B0019 | 0012 | 0102 | --- |
| B0020 | 0022 | 0018 | --- |

1/ Category B connectors are for original installation only. They will not be stocked or acquired by the Government.



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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-39012, this document references the following:

FED-STD-H28
MIL-STD-202-101
MIL-STD-202-106
MIL-STD-202-107
MIL-STD-202-204
MIL-STD-202-213
MIL-STD-202-301
MIL-STD-202-302
MIL-STD-348
SAE-AS22520
SAE-AS39029

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force – 85
NASA – NA
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-2020-030)

Review activities:
Army - AT, AV, EA, MI
Navy - AS, MC, OS, SH
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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Check the source to verify that this is the current version before use.